Hike: Processing analytics queries 20X faster with Google Cloud

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About Hike

Hike is an Indian AI-led Unicorn startup committed to building a new social future and believes this future will be defined by "joyful" products built around people. The business is targeting social niches through two unique products, Hike Sticker Chat and WinZO. Hike Sticker Chat is an "expression" platform that aims to reduce individuals' dependence on the keyboard, while WinZO—in which Hike has led an investment—is an Indian online arcade platform. In August 2016, Hike became the tenth Indian startup to be valued at over $1 billion. The

provided by a nev
based messaging
Sticker Chat.

Google Cloud results

- Processes analytics queries 20X faster than previously
- Doubles compute throughput
- Uses Google Cloud Machine Learning Engine managed, distributed capabilities to train complex models on TensorFlow that provide delightful local sticker recommendations through Hike Sticker Chat

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India is a market of opportunity
provide messaging apps to cons
than 1.3 billion people, the coun

business has raised over $261 million in capital from top-tier investors such as SoftBank, Tiger Global, Tencent, Foxconn, and Bharti.

Industries: Technology

Location: India

, BigQuery

About CloudCover

A Google Cloud Partner, CloudCover builds, automates and manages systems at cloud scale. The business describes itself as delivering the insane potential of the public cloud to start-ups and agile enterprises through a combination of weaponized geekiness, extreme automation, and battle-scarred experience.

Google Cloud (https://cloud.google.com/)

Cloud Storage (https://cloud.google.com/storage/)

Dataflow (https://cloud.google.com/dataflow/)

Cloud Dataproc (https://cloud.google.com/dataproc/)

BigQuery (https://cloud.google.com/bigquery/)

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messaging app providers face a
challenge from Hike (https://hike.
internet and technology startup.
Hike provides innovative produc
Messenger and more recently th
learning-enabled Hike Sticker Cha
enables young people in the country to express themselves through digital stickers.

Compute Engine (https://cloud.google.com/compute/)

AI Platform (https://cloud.google.com/ai-platform/)

Cloud Load Balancing (https://cloud.google.com/load-balancing/)

TensorFlow (https://www.tensorflow.org/)

The business says it understands the people of India and communication like no one else, while its mission is to reduce individuals' dependency on the keyboard. To do this, Hike is building one of the largest repositories of AI and machine-learning-enabled stickers for Hike Sticker Chat. This messaging platform is, according to Hike, the only product of its type that enables conversations through stickers covering more than 40 languages and local dialects.

Founded by Kavin Bharti Mittal, the Delhi-based venture is backed by SoftBank, Tencent, Tiger Global, Foxconn, and Bharti. To date, Hike has raised $261 million in funding. In August 2016, Hike raised its Series D round of funding, led by Tencent and Foxconn, at a valuation of $1.4 billion. The business is one of the fastest Indian startups to achieve Unicorn status, doing so in less than four years.

Hike started operations on a multinational cloud service. However, as user numbers and usage grew, the business began exploring options to improve performance and stability, reduce costs, and cut administration loads. In particular, Hike wanted to reduce latency between cloud data centers.

"We chose Google Cloud because of its very broad set of services and features."

—Aditya Gupta, Director, Engineering, Hike

Focus on product development

"We aimed to move away from a technology stack with single points of failure to a horizontally scaled, highly reliable, distributed systems and managed services environment that enabled us to focus on product development rather than operations," says Aditya Gupta, Director, Engineering, Hike.

Hike then began exploring the opportunities presented by Google Cloud (https://cloud.google.com/). The business held a number of executive-level meetings with Google to understand the capabilities, roadmap, and track record of the cloud service. It then decided to proceed with a proof of concept with Google Cloud Partner CloudCover (https://cldcvr.com/).

The proof of concept revealed that when Cloud Load Balancing (https://cloud.google.com/load-balancing/) was operating, latency between the Google Cloud data center in Taiwan and Delhi, India, was less than the latency between the incumbent cloud provider's data center and Delhi. Further, compute throughput was up to two times greater on Compute Engine (https://cloud.google.com/compute/) than on the equivalent service, while Hike could complete more then 1 million connections on Compute Engine – up from 500,000 connections on the incumbent service.

Migrate to Google Cloud

The success of the exercise prompted Hike to migrate its messaging app to Google Cloud. "We chose Google Cloud because of its very broad set of services and features," explains Gupta. "In addition, Google's innovation mindset and the richness of the partnership would allow us to be onboarded quickly to machine learning services such as Cloud Machine Learning Engine (https://cloud.google.com/ml-engine/)."

The business called on Google Cloud Professional Services (https://cloud.google.com/consulting/) (Technical Account Management) (https://cloud.google.com/tam/) to help ensure a seamless lift-and-shift migration over two months. Google Cloud Professional Services initially undertook a technical infrastructure kickoff to establish a foundation for architecture requirements such as identity and access management and security.

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>
> *—Aditya Gupta, Director, Engineering, Hike*

Google Cloud Professional Services team delivers smooth migration

Google Cloud Professional Services worked closely with Hike to map out and deliver the Google Cloud architecture that would deliver the greatest value to the business. The Professional Services team also worked with Hike to resolve product and support queries quickly; provided project background for product and support teams; and organized project meetings and early adopter program access.

In addition, Professional Services team members worked on site at least once a week, coordinated external support during critical migration periods, and coordinated teams in five countries for a single, 17-hour migration marathon. Over 60 days, the business migrated 7,000 processor cores, running virtual machine instances used for messaging infrastructure and analytics, to Google Cloud.

Throughout the exercise, Google Cloud Professional Services worked with CloudCover to educate the customers' technology teams to achieve proficiency with Google Cloud. The teams soon built up skills and knowledge of best practices and began applying them to the Google Cloud environment.

The Hike Google Cloud architecture comprises virtual machine instances running in Compute Engine; Cloud Storage (https://cloud.google.com/storage/) for unified object storage; networking; a BigQuery (https://cloud.google.com/bigquery/) analytics data warehouse; Cloud Dataflow (https://cloud.google.com/dataflow/) to transform and enrich data; Cloud Load Balancing to distribute workloads to maximize efficiency; and Cloud

Dataproc (https://cloud.google.com/dataproc/) to run Hadoop clusters.

Hike is also stepping up its AI & machine learning capabilities. It uses Google Cloud Machine Learning Engine managed, distributed computing capabilities to train complex models on TensorFlow (https://www.tensorflow.org/). This powers key use cases such as delightful local sticker recommendations on Hike Sticker Chat. Hike is also investing heavily (https://ai.hike.in/) on AI and machine learning research.

"Google Cloud has played an important role in enabling us to continue to innovate and realize our mission of reducing dependency on the keyboard."

—Aditya Gupta, Director, Engineering, Hike

Hike has achieved a range of benefits from its Google Cloud deployment. As well as reduced latency, improved compute throughput, and increased connection handling, Google Cloud managed services have enabled the business to reduce the time and effort required to administer core infrastructure, with the saved resources allocated to improving its messaging product.

"Managed services are beginning to reduce our operational overheads," says Gupta. "For example, managed instance groups and Cloud Load Balancing are reducing our instance count and costs, thereby reducing involvement from DevOps and developer teams."

Google Cloud 20% cheaper

Gupta and his team have calculated that running for three years on Google Cloud will cost, including the cost of migration, 20 percent less than on its previous platform. BigQuery is processing queries 20 times faster than a similar service offered by the previous provider, while storing 125TB of data and streaming 1.5TB of data daily. Furthermore, Hike's analytics pipeline costs 80 percent less than in its previous environment.

"Google Cloud has played an important role in enabling us to continue to innovate and realize our mission of reducing dependency on the keyboard," says Gupta.

Learn More

Reach out to our team to see how Google Cloud can help your business

https://cloud.google.com